Day 2: Python Operators (Detailed Notes for Data Analysis Students)

WHAT ARE OPERATORS?

Operators are special symbols used to perform operations on variables and values.

EXAMPLE

```
a = 10
b = 5

print(a + b)
```

Output:

```
15
```

Here, `+` is an operator.

Types of Operators in Python

1. Arithmetic Operators
2. Comparison Operators
3. Logical Operators
4. Assignment Operators
5. Membership Operators
6. Identity Operators
7. Bitwise Operators

For Data Analysts, the first 4 are most important.

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1. Arithmetic Operators

Used for mathematical calculations.

Operator	Meaning	Example
+	Addition	10 + 5

-	Subtraction	10 - 5
*	Multiplication	10 * 5
/	Division	10 / 5
//	Floor Division	10 // 3
%	Modulus	10 % 3
**	Power	2 ** 3

ADDITION

```
sales_jan = 50000
sales_feb = 45000

total_sales = sales_jan + sales_feb

print(total_sales)
```

Output:

```
95000
```

SUBTRACTION

```
sales = 50000
cost = 35000

profit = sales - cost

print(profit)
```

Output:

```
15000
```

MULTIPLICATION

```
price = 250
quantity = 10

amount = price * quantity

print(amount)
```

Output:

```
2500
```

DIVISION

```
total_sales = 100000
employees = 5

avg_sales = total_sales / employees

print(avg_sales)
```

Output:

```
20000.0
```

MODULUS (%)

Returns remainder.

```
print(10 % 3)
```

Output:

```
1
```

EVEN/ODD CHECK

```
number = 12

print(number % 2)
```

Output:

```
0
```

If remainder = 0 → Even Number

POWER OPERATOR

```
print(5 ** 2)
```

Output:

```
25
```

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2. Comparison Operators

Used to compare values.

Result is always:

- True
- False

Operator	Meaning
==	Equal
!=	Not Equal
>	Greater Than
<	Less Than
>=	Greater Than Equal
<=	Less Than Equal

EQUAL TO

```python
a = 10
b = 10

print(a == b)
```

Output:

```
True
```

NOT EQUAL TO

```python
a = 10
b = 20

print(a != b)
```

Output:

```
True
```

GREATER THAN

```python
salary = 50000

print(salary > 30000)
```

Output:

```
True
```

REAL DATA ANALYSIS EXAMPLE

```python
sales = 75000

print(sales > 50000)
```

Output:

```
True
```

Used for KPI checks.

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3. Logical Operators

Used to combine conditions.

Operator	Meaning
and	Both conditions must be True
or	Any one condition True
not	Reverse result

AND OPERATOR

```
salary = 50000
experience = 3

print(salary > 30000 and experience >= 2)
```

Output:

```
True
```

OR OPERATOR

```
salary = 25000
experience = 5

print(salary > 30000 or experience >= 5)
```

Output:

```
True
```

NOT OPERATOR

```
print(not True)
```

Output:

```
False
```

4. Assignment Operators

Used to assign values.

Operator	Example
=	x = 10
+=	x += 5
-=	x -= 5
*=	x *= 5
/=	x /= 5

EXAMPLE

```
salary = 10000

salary += 5000

print(salary)
```

Output:

```
15000
```

Business Examples

PROFIT CALCULATION

```
sales = 120000
cost = 80000

profit = sales - cost

print("Profit =", profit)
```

Output:

```
Profit = 40000
```

DISCOUNT CALCULATION

```
price = 5000

discount = price * 10 / 100

final_price = price - discount

print(final_price)
```

Output:

```
4500
```

AVERAGE SALES

```
jan = 50000
feb = 60000
mar = 70000

average = (jan + feb + mar) / 3

print(average)
```

Output:

```
60000
```

Mini Project 1: Salary Bonus Calculator

```
salary = float(input("Enter Salary: "))

bonus = salary * 10 / 100

final_salary = salary + bonus

print("Bonus =", bonus)
print("Final Salary =", final_salary)
```

Mini Project 2: Student Percentage

```
maths = 80
science = 90
english = 85

percentage = (maths + science + english) / 3

print("Percentage =", percentage)
```

Mini Project 3: Profit Margin

```
sales = 100000
cost = 70000

profit = sales - cost

profit_margin = (profit / sales) * 100

print("Profit Margin =", profit_margin)
```

Practice Questions

ARITHMETIC OPERATORS

1. Add two numbers.
2. Subtract two numbers.
3. Multiply two numbers.
4. Divide two numbers.
5. Find remainder of 25 ÷ 4.

6. Find square of 15.
7. Find cube of 5.
8. Calculate total sales of 3 months.
9. Calculate average marks.
10. Calculate total salary of 5 employees.

COMPARISON OPERATORS

11. Check whether 50 is greater than 30.
12. Check whether 20 equals 25.
13. Compare two salaries.
14. Check whether age is above 18.
15. Compare sales with target.

LOGICAL OPERATORS

16. Check if salary > 30000 AND experience > 2.
17. Check if sales > target OR profit > target.
18. Use NOT operator with True.
19. Create a login condition.
20. Create a bonus eligibility condition.

BUSINESS PROBLEMS

21. Calculate employee bonus.
22. Calculate GST (18%).
23. Calculate discount (15%).
24. Calculate profit.
25. Calculate profit margin.
26. Calculate average sales.
27. Calculate yearly salary.
28. Calculate monthly EMI.
29. Calculate electricity bill.
30. Calculate fuel cost for a trip.

Assignment (Data Analyst Level)

Create a program that:

1. Takes Sales Amount
2. Takes Cost Amount
3. Calculates Profit
4. Calculates Profit %
5. Checks whether Profit % is greater than 20%

Example:

```
Sales: 100000
Cost: 70000

Profit: 30000
Profit %: 30

Target Achieved: True
```

INTERVIEW QUESTIONS

1. What is an operator?
2. Difference between `=` and `==`?
3. Difference between `/` and `//`?
4. What is modulus operator?
5. What is logical AND?
6. What is logical OR?
7. What is the use of comparison operators?
8. What are assignment operators?
9. How is modulus used to find even/odd numbers?
10. Give a real-life business example of operators.